A Unified Readout–Compression Framework

Atemporal Carriers, Spacetime Realizations, Parity Persistence,
and a Sterile-Neutrino Tachyonic Driver

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Abstract

We present a unified formal framework in which a spacetime-dependent manifestation is modeled as the temporal projection of an atemporal invariant carrier. The main structural principle is that collapse of manifest readout does not imply annihilation of the underlying carrier, provided compression is admissible and parity-compatible. The framework is extended by introducing a sterile-neutrino driver with two tachyonic regimes: a slow cohesive mode and a fast expansive mode. These are treated as complementary dynamical channels acting on the spacetime realization while preserving latent support in the carrier. The central theorem establishes persistence of the atemporal carrier under readout collapse, together with a parity-resolved base–residue decomposition of the compressed state.

1 Introduction

Many formal structures distinguish poorly between what is *manifestly readable* and what is merely *latently present*. This distinction becomes essential when an observable output vanishes while the underlying organization remains nontrivial.

We introduce a minimal framework based on the following ingredients:

(i) an atemporal invariant carrier,

(ii) a spacetime realization obtained by projection,

(iii) a manifest readout channel,

(iv) an admissible compression operator,

(v) a latent encoding preserving structure beyond visible output,

(vi) a parity-preserving decomposition into base and residue channels,

(vii) a sterile-neutrino tachyonic driver with slow and fast regimes.

The basic structural claim is:

$$\boxed{\Sigma_{\mathcal{M}} = \Pi_{\mathcal{M}}(\Sigma_*),}$$

where Σ_* is an atemporal carrier and $\Sigma_{\mathcal{M}}$ is its spacetime realization.

The basic dynamical claim is:

$$\boxed{\mathcal{R}(\mathcal{C}(\Sigma_{\mathcal{M}})) = 0 \;\not\Rightarrow\; \Sigma_* = 0.}$$

2 Basic Setting

Let $\mathcal{M}$ be a spacetime manifold and let $\mathcal{X}(\mathcal{M})$ denote a space of admissible fields, structured trajectories, or state realizations over $\mathcal{M}$.

Definition 1 (Atemporal carrier)**.** *An* atemporal carrier *is an object*

$$\Sigma_* = (\mathcal{I}, Q),$$

where:

(a) $\mathcal{I}$ is an invariant identity core,

(b) $Q = \{q_\alpha\}_{\alpha \in A}$ is an internal configuration bundle.

The carrier Σ_* is not parametrized by time.

Definition 2 (Spacetime realization)**.** *A* spacetime realization *is an element*

$$\Sigma_{\mathcal{M}} \in \mathcal{X}(\mathcal{M})$$

obtained via a realization map

$$\Pi_{\mathcal{M}} : \Sigma_* \mapsto \Sigma_{\mathcal{M}}.$$

Definition 3 (Manifest readout and latent encoding)**.** *A* manifest readout *is a map*

$$\mathcal{R} : \mathcal{X}(\mathcal{M}) \to \mathcal{V},$$

where $\mathcal{V}$ is a value space. A latent encoding *is a map*

$$\mathcal{L} : \mathcal{X}(\mathcal{M}) \to \mathcal{W},$$

where $\mathcal{W}$ is an auxiliary representation space retaining structure not necessarily visible in $\mathcal{V}$.

Definition 4 (Compression operator)**.** *A* compression operator *is a map*

$$\mathcal{C} : \mathcal{X}(\mathcal{M}) \to \mathcal{X}(\mathcal{M})$$

which may reduce manifest amplitude while preserving latent support.

3 Admissibility and Parity Structure

Definition 5 (Admissible compression)**.** *A compression operator $\mathcal{C}$ is* admissible *if*

$$\mathcal{R}(\mathcal{C}(x)) = 0 \quad \not\Rightarrow \quad \mathcal{L}(\mathcal{C}(x)) = 0$$

for structurally relevant $x \in \mathcal{X}(\mathcal{M})$.

Definition 6 (Parity-resolved realization)**.** *A realization $x \in \mathcal{X}(\mathcal{M})$ is called* parity-resolved *if there exists an involution P on $\mathcal{X}(\mathcal{M})$ such that*

$$x = x_{\text{even}} + x_{\text{odd}},$$

with

$$x_{\text{even}} = \frac{1}{2}(I + \mathsf{P})x, \qquad x_{\text{odd}} = \frac{1}{2}(I - \mathsf{P})x.$$

Definition 7 (Parity-compatible compression)**.** *A compression operator $\mathcal{C}$ is* parity-compatible *if*

$$\mathcal{C}(x) = \mathcal{C}(x_{\text{even}}) + \mathcal{C}(x_{\text{odd}})$$

and does not identify the even and odd sectors.

This is the structural analogue of the even/odd split in parity-filtered expansions: one channel supplies a stable base, the other a structured residual correction.

4 Base and Residue Channels

Definition 8 (Base and residue)**.** *For a parity-resolved realization x, define*

$$x_{\text{base}} := x_{\text{even}}, \qquad x_{\text{res}} := x_{\text{odd}}.$$

After compression,

$$\mathcal{C}(x) = \mathcal{C}(x_{\text{base}}) + \mathcal{C}(x_{\text{res}}).$$

We refer to these as the base channel *and the* residue channel*.*

The intended interpretation is:

- the base channel carries stable transmissive content,
- the residue channel carries parity-sensitive correction content.

5 Sterile-Neutrino Tachyonic Driver

We now introduce the dynamical driver acting on the spacetime realization.

Definition 9 (Sterile-neutrino driver)**.** *A* sterile-neutrino driver *is a field or effective mode N_s coupled to the spacetime realization through a tachyonic scalar parameter ϕ, with effective state parameter*

$$w = \dot{\phi}^2 - 1.$$

This produces two distinguished regimes:

Definition 10 (Slow and fast tachyonic modes)**.** *The driver is said to be in:*

(i) a slow tachyonic mode *if $\dot{\phi}^2 \ll 1$, so that $w \approx -1$,*

(ii) a fast tachyonic mode *if $\dot{\phi}^2 \to 1$, so that $w \to 0$.*

The slow mode acts as a cohesive channel; the fast mode acts as an expansive or transport channel.

Definition 11 (Dual driver decomposition)**.** *Let $\mathcal{D}_N$ denote the sterile-neutrino driver acting on $\Sigma_{\mathcal{M}}$. We write*

$$\mathcal{D}_N = \mathcal{D}_{\text{coh}} + \mathcal{D}_{\text{exp}},$$

where $\mathcal{D}_{\text{coh}}$ is the slow cohesive mode and $\mathcal{D}_{\text{exp}}$ is the fast expansive mode.

These two regimes will be treated as complementary dynamical channels rather than contradictory ones.

6 Projection Principle

Axiom 1 (Projection principle)**.** *Every admissible spacetime realization is obtained as a temporal projection of an atemporal carrier:*

$$\Sigma_{\mathcal{M}} = \Pi_{\mathcal{M}}(\Sigma_*).$$

Axiom 2 (Latent persistence)**.** *If $\Sigma_{\mathcal{M}}$ undergoes admissible compression, then latent support may remain nontrivial even when manifest readout vanishes:*

$$\mathcal{R}(\mathcal{C}(\Sigma_{\mathcal{M}})) = 0 \quad \not\Rightarrow \quad \mathcal{L}(\mathcal{C}(\Sigma_{\mathcal{M}})) = 0.$$

Axiom 3 (Parity persistence)**.** *Parity decomposition survives admissible compression.*

Axiom 4 (Dual-driver compatibility)**.** *The cohesive and expansive components of the sterile-neutrino driver act on the realization without annihilating the atemporal carrier.*

7 Lemmas

Lemma 1 (Readout collapse is channel-relative)**.** *For $x \in \mathcal{X}(\mathcal{M})$, the condition*

$$\mathcal{R}(x) = 0$$

concerns only the manifest output channel and does not by itself imply structural triviality of x.

Proof. By construction, $\mathcal{R}$ maps only into the visible value space $\mathcal{V}$. Simultaneously, $\mathcal{L}(x)$ may remain nonzero in $\mathcal{W}$. Therefore vanishing of $\mathcal{R}(x)$ does not force total collapse of structure. □

Lemma 2 (Parity survives admissible compression)**.** *Let $x = x_{\text{even}} + x_{\text{odd}}$ and let $\mathcal{C}$ be parity-compatible. Then the compressed realization retains a nontrivial decomposition*

$$\mathcal{C}(x) = \mathcal{C}(x_{\text{even}}) + \mathcal{C}(x_{\text{odd}}),$$

with the two sectors structurally distinct.

Proof. This follows directly from parity-compatibility. □

Lemma 3 (Latent residue persists under admissible compression)**.** *If $x_{\text{odd}} \neq 0$ structurally and $\mathcal{C}$ is admissible, then*

$$\mathcal{L}(\mathcal{C}(x_{\text{odd}})) \neq 0$$

throughout the admissible regime.

Proof. Admissibility states that collapse of manifest readout does not imply collapse of the latent channel. Since the odd sector remains structurally separate after compression, its latent image remains nontrivial. □

Lemma 4 (Dual driver changes exposure, not carrier support)**.** *Let $\mathcal{D}_N = \mathcal{D}_{\text{coh}} + \mathcal{D}_{\text{exp}}$ be the sterile-neutrino driver. Then transitions between slow and fast tachyonic regimes may alter the visible realization without forcing annihilation of the atemporal carrier.*

Proof. The driver acts on the spacetime realization $\Sigma_{\mathcal{M}}$, not directly on the atemporal carrier Σ_*. Since $\Sigma_{\mathcal{M}}$ is a projection of Σ_*, changes in dynamical regime modify the manifestation channel and its compression/readout behavior, but do not by themselves imply vanishing of the source carrier. □

8 Main Theorem

Theorem 1 (Carrier persistence under readout collapse and dual driving)**.** *Let $\Sigma_* = (\mathcal{I}, Q)$ be an atemporal carrier and let*

$$\Sigma_{\mathcal{M}} = \Pi_{\mathcal{M}}(\Sigma_*)$$

be its spacetime realization. Assume:

(i) $\mathcal{C}$ is admissible,

(ii) $\Sigma_{\mathcal{M}}$ is parity-resolved,

(iii) $\mathcal{C}$ is parity-compatible,

(iv) $\Sigma_{\mathcal{M}}$ is driven by a sterile-neutrino tachyonic operator

$$\mathcal{D}_N = \mathcal{D}_{\text{coh}} + \mathcal{D}_{\text{exp}}.$$

If

$$\mathcal{R}(\mathcal{C}(\mathcal{D}_N(\Sigma_{\mathcal{M}}))) = 0,$$

then this does not imply $\Sigma_* = 0$*. More precisely, manifest collapse of the driven realization is compatible with nontrivial latent persistence of the atemporal carrier, together with a surviving parity-resolved base–residue structure.*

Proof. Write the realization in parity-resolved form:

$$\Sigma_{\mathcal{M}} = \Sigma_{\text{even}} + \Sigma_{\text{odd}}.$$

Applying the dual sterile-neutrino driver gives

$$\mathcal{D}_N(\Sigma_{\mathcal{M}}) = \mathcal{D}_{\text{coh}}(\Sigma_{\mathcal{M}}) + \mathcal{D}_{\text{exp}}(\Sigma_{\mathcal{M}}).$$

This changes the realization dynamically, but by Lemma 7.4 does not by itself annihilate the carrier.

Now compress the driven realization:

$$\mathcal{C}(\mathcal{D}_N(\Sigma_{\mathcal{M}})).$$

Since $\mathcal{C}$ is parity-compatible,

$$\mathcal{C}(\mathcal{D}_N(\Sigma_{\mathcal{M}})) = \mathcal{C}(\mathcal{D}_N(\Sigma_{\text{even}})) + \mathcal{C}(\mathcal{D}_N(\Sigma_{\text{odd}})).$$

Thus the compressed state still admits a base–residue split.

Suppose now that

$$\mathcal{R}(\mathcal{C}(\mathcal{D}_N(\Sigma_{\mathcal{M}}))) = 0.$$

By Lemma 7.1, this is only a statement about the manifest channel. It does not imply total structural collapse.

By Lemma 7.2, parity survives admissible compression, so the even and odd sectors remain structurally distinguishable. Hence the compressed state still possesses a base channel and a residue channel.

By Lemma 7.3, the odd sector, if structurally nontrivial before compression, retains a nonzero latent image:

$$\mathcal{L}(\mathcal{C}(\mathcal{D}_N(\Sigma_{\text{odd}}))) \neq 0.$$

Therefore the residue channel persists latently even if the visible readout has vanished.

Since $\Sigma_{\mathcal{M}}$ is by assumption the spacetime projection of Σ_*, complete annihilation of Σ_* would force triviality of all admissible projections and their latent encodings. But the latent encoding remains nontrivial. Hence

$$\Sigma_* \neq 0.$$

Therefore manifest collapse of the driven realization does not imply annihilation of the atemporal carrier. It indicates a transition from visible exposure into a compressed latent regime, with parity architecture preserved. □

9 Corollaries

Corollary 1 (Projection formula)**.** *Every admissible realization is a temporal projection of an atemporal carrier:*

$$\boxed{\Sigma_{\mathcal{M}} = \Pi_{\mathcal{M}}(\Sigma_*).}$$

Corollary 2 (Collapse does not imply carrier loss)**.** *In the admissible dual-driver regime,*

$$\boxed{\mathcal{R}(\mathcal{C}(\mathcal{D}_N(\Sigma_{\mathcal{M}}))) = 0 \;\not\Rightarrow\; \Sigma_* = 0.}$$

Corollary 3 (Base–residue survival)**.** *The compressed driven realization admits the split*

$$\mathcal{C}(\mathcal{D}_N(\Sigma_{\mathcal{M}})) = (\textit{base channel}) + (\textit{residue channel}),$$

where the residue channel remains latently supported.

Corollary 4 (Cohesive and expansive complementarity)**.** *The slow and fast tachyonic regimes are compatible as dual dynamical channels acting on the same spacetime realization.*

10 Interpretive Compression

The framework may be compressed into the following diagram:

$$\Sigma_* \xrightarrow{\Pi_{\mathcal{M}}} \Sigma_{\mathcal{M}} \xrightarrow{\mathcal{D}_N} \mathcal{D}_N(\Sigma_{\mathcal{M}}) \xrightarrow{\mathcal{C}} \mathcal{C}(\mathcal{D}_N(\Sigma_{\mathcal{M}}))$$

with

$$\mathcal{R}(\mathcal{C}(\mathcal{D}_N(\Sigma_{\mathcal{M}}))) = 0 \qquad \text{but} \qquad \mathcal{L}(\mathcal{C}(\mathcal{D}_N(\Sigma_{\mathcal{M}}))) \neq 0.$$

The structural reading is therefore:

manifest output may collapse, while latent support persists under projection, compression, parity splitting, and dual driving.

11 Conclusion

We have presented a unified framework combining:

- an atemporal invariant carrier,
- a spacetime realization map,
- admissible compression,
- manifest readout and latent encoding,
- parity-preserving base–residue decomposition,
- a sterile-neutrino tachyonic driver with slow cohesive and fast expansive modes.

The main theorem shows that collapse of visible output does not imply annihilation of the atemporal carrier. Instead, the system enters a compressed latent regime in which parity architecture survives and the residue channel remains structurally supported.

The final structural verdict is:

$$\boxed{\text{Readout collapse changes accessibility, not carrier support.}}$$

12 Addendum: Monodromy, Branch Switching, and Closure Completion

This addendum records two structural extensions of the main framework. The first concerns branch switching under phase monodromy; the second concerns globally completed readout beyond any single finite projection.

12.1 Monodromic phase switching

The phase-sensitive part of the framework may be sharpened by recalling that branch-valued objects are not globally single-valued on $\mathbb{C}^\times$. Let

$$z = re^{i\theta}, \qquad r > 0, \quad \theta \in \mathbb{R}.$$

The principal square-root branch is given by

$$\sqrt{re^{i\varphi}} = \sqrt{r}\, e^{i\varphi/2}, \qquad \varphi \in (-\pi, \pi].$$

Under one full positive loop around the origin,

$$\theta \mapsto \theta + 2\pi,$$

the square-root transforms as

$$\sqrt{r}\, e^{i\theta/2} \mapsto \sqrt{r}\, e^{i(\theta+2\pi)/2} = \sqrt{r}\, e^{i\theta/2} e^{i\pi} = -\sqrt{r}\, e^{i\theta/2}.$$

Hence the monodromy factor is

$$e^{i\pi} = -1.$$

Proposition 1 (Monodromic branch switch)**.** *A full 2π-rotation in the base phase induces a π-shift in the square-root phase, thereby switching sheets through the factor $e^{i\pi} = -1$.*

Proof. The argument increment 2π is halved by the square-root map, producing a phase increment of π. The exponential factor is therefore $e^{i\pi} = -1$, which exchanges the two sheets. □

This motivates the interpretation of π as a *monodromic switch value*: not merely a geometric constant, but the exact half-period that flips the branch sheet.

12.2 Closure-completed readout

We now refine the readout architecture by introducing a three-layer decomposition:

$$\mathcal{H}_{\text{tot}} = \mathcal{H}_B \otimes \mathcal{H}_H \otimes \mathcal{H}_T,$$

where:

- $\mathcal{H}_B$ is the bulk or directly readable sector,
- $\mathcal{H}_H$ is a boundary-memory sector,
- $\mathcal{H}_T$ is an auxiliary phase sector.

Let $\rho_B(t) \in D(\mathcal{H}_B)$, $\rho_H(t) \in D(\mathcal{H}_H)$, and $\rho_T(t) \in D(\mathcal{H}_T)$ denote the corresponding states. Define a readout projection

$$P_\Theta : D(\mathcal{H}_B) \to Y,$$

an auxiliary phase rotation

$$R_\phi : \mathcal{H}_T \to \mathcal{H}_T,$$

and an adjoint reconstruction map

$$A^* : B(\mathcal{H}_H \otimes \mathcal{H}_T) \to B(\mathcal{H}_B).$$

The completed observable output is then

$$Y_{\text{full}}(t) = P_\Theta(\rho_B(t) + A^*(\rho_H(t) \otimes \rho_T(t)))\,,$$

whereas the bare bulk readout is

$$Y_{\text{bulk}}(t) = P_\Theta(\rho_B(t)).$$

Definition 12 (Closure defect)**.** *The* closure defect *is the quantity*

$$\Delta_\Theta(t) := Y_{\text{full}}(t) - Y_{\text{bulk}}(t).$$

Definition 13 (Closed readout)**.** *The readout is said to be* closed *on a class of states if*

$$\Delta_\Theta(t) = 0$$

for all relevant states and times.

Definition 14 (Closure gap)**.** *The system exhibits a* closure gap *if there exist states and times such that*

$$\Delta_\Theta(t) \neq 0.$$

Proposition 2 (Finite readout need not be globally complete)**.** *A finite bulk readout may fail to certify the latent global structure, even when the latter is internally coherent.*

Proof. The map P_Θ is not assumed injective. Hence distinct latent completions may yield the same bulk output. The reconstructed contribution $A^*(\rho_H(t) \otimes \rho_T(t))$ can therefore change the completed readout without being visible in the bare bulk channel alone. Thus $Y_{\text{bulk}}(t)$ need not be globally complete. □

12.3 Compatibility with the main framework

The closure-completed readout is compatible with the main carrier-realization architecture by identifying the manifest realization with the bulk channel and the latent persistence with the reconstructed completion. In particular, the observable output is no longer determined by the bulk sector alone, but by the completed expression

$$P_\Theta(\rho_B(t) + A^*(\rho_H(t) \otimes \rho_T(t)))\,.$$

The monodromic branch switch is then interpreted as a special phase-frame event: a full loop in the base phase may alter the observable branch of the realization even when the latent carrier remains unchanged. Thus branch switching changes readout frame, not carrier support.

Corollary 5 (Monodromy-completion principle)**.** *A phase-induced branch switch may alter the observable realization without implying any annihilation of the underlying latent carrier.*

Proof. Monodromy acts through branch-sensitive phase structure at the level of realization and readout. By the main theorem of the paper, collapse or alteration of manifest output does not imply collapse of the atemporal carrier. Hence branch switching is a readout-level effect. □

A Appendix: Dry Axiomatic Skeleton

For reference, the completed readout architecture may be summarized by the following dry axioms:

Axiom 1. There exists a finite observable projection

$$P_\Theta : D(\mathcal{H}_B) \to Y$$

which need not be injective.

Axiom 2. There exists a time-dependent boundary-memory family

$$\rho_H(t) \in D(\mathcal{H}_H).$$

Axiom 3. There exists an auxiliary phase sector $\mathcal{H}_T$ with nontrivial phase action

$$R_\phi : \mathcal{H}_T \to \mathcal{H}_T.$$

Axiom 4. There exists an adjoint reconstruction map

$$A^* : B(\mathcal{H}_H \otimes \mathcal{H}_T) \to B(\mathcal{H}_B).$$

Axiom 5. The completed output is

$$Y_{\text{full}}(t) = P_\Theta(\rho_B(t) + A^*(\rho_H(t) \otimes \rho_T(t)))\,.$$

The resulting gap

$$\Delta_\Theta(t) = Y_{\text{full}}(t) - Y_{\text{bulk}}(t)$$

measures the difference between finite local readout and globally completed structure.

B The Unified Readout–Persistence Spine

We now promote the completed readout architecture to the central structural layer of the framework.

B.1 Unified readout data

Let

$$\Sigma_* = (\mathcal{I}, Q)$$

be an atemporal invariant carrier, and let

$$\Pi_\mathcal{M} : \Sigma_* \mapsto \Sigma_\mathcal{M}$$

be its spacetime realization map.

Let

$$\mathcal{D}_N = \mathcal{D}_{\text{coh}} + \mathcal{D}_{\text{exp}}$$

be the sterile-neutrino tachyonic driver, with slow cohesive and fast expansive modes.

Let

$$P_\Theta : D(\mathcal{H}_B) \to Y$$

be the readout projection,

$$A^* : B(\mathcal{H}_H \otimes \mathcal{H}_T) \to B(\mathcal{H}_B)$$

the adjoint reconstruction map, and

$$R_\phi : \mathcal{H}_T \to \mathcal{H}_T$$

the auxiliary phase rotation.

Let $\rho_H(t) \in D(\mathcal{H}_H)$ and $\rho_T(t) \in D(\mathcal{H}_T)$ denote the boundary-memory and phase-sector states.

Definition 15 (Unified readout value)**.** *The completed readout value of the carrier Σ_* at parameter Θ is defined by*

$$Y_{RV}(\Sigma_*; \Theta) := P_\Theta(\mathcal{D}_N(\Pi_\mathcal{M}(\Sigma_*)) + A^*(\rho_H(t) \otimes R_\phi \rho_T(t)))\,(-1)^m.$$

The factor $(-1)^m$ encodes the monodromic branch switch in the readout frame.

B.2 Closure-completed structure

The bare driven realization is

$$Y_{\text{bare}}(t) = P_\Theta(\mathcal{D}_N(\Pi_\mathcal{M}(\Sigma_*))),$$

while the completed readout includes the reconstructed boundary-phase contribution:

$$Y_{RV}(\Sigma_*;\Theta) = P_\Theta(\mathcal{D}_N(\Pi_\mathcal{M}(\Sigma_*)) + A^*(\rho_H(t) \otimes R_\phi \rho_T(t)))\,(-1)^m.$$

Definition 16 (Closure defect)**.** *The closure defect is*

$$\Delta_\Theta(t) := Y_{RV}(\Sigma_*;\Theta) - Y_{\text{bare}}(t).$$

Thus $\Delta_\Theta(t) \neq 0$ measures the gap between finite local readout and closure-completed structure.

B.3 Parity-resolved decomposition

Assume the spacetime realization is parity-resolved:

$$\Sigma_\mathcal{M} = \Sigma_{\text{even}} + \Sigma_{\text{odd}},$$

with

$$\Sigma_{\text{even}} = \frac{1}{2}(I + \mathsf{P})\Sigma_\mathcal{M}, \qquad \Sigma_{\text{odd}} = \frac{1}{2}(I - \mathsf{P})\Sigma_\mathcal{M}.$$

Under admissible parity-compatible compression C, we obtain

$$C(\mathcal{D}_N(\Sigma_\mathcal{M})) = C(x_{\text{base}}) + C(x_{\text{res}}),$$

where

$$x_{\text{base}} := \Sigma_{\text{even}}, \qquad x_{\text{res}} := \Sigma_{\text{odd}}.$$

The even sector is the stable base channel; the odd sector is the parity-sensitive residue channel.

B.4 MASTER RV Theorem v3

Theorem 2 (Unified Carrier–Readout Persistence Theorem)**.** *Let $\Sigma_* = (\mathcal{I}, Q)$ be an atemporal invariant carrier and let*

$$\Sigma_\mathcal{M} = \Pi_\mathcal{M}(\Sigma_*)$$

be its spacetime realization. Assume:

(i) C is an admissible compression operator,

(ii) $\Sigma_\mathcal{M}$ is parity-resolved,

(iii) C is parity-compatible,

(iv) the realization is driven by

$$\mathcal{D}_N = \mathcal{D}_{\text{coh}} + \mathcal{D}_{\text{exp}},$$

(v) the completed readout is given by

$$Y_{RV}(\Sigma_*;\Theta) = P_\Theta(\mathcal{D}_N(\Pi_\mathcal{M}(\Sigma_*)) + A^*(\rho_H(t) \otimes R_\phi \rho_T(t)))\,(-1)^m.$$

If

$$R(C(\mathcal{D}_N(\Pi_{\mathcal{M}}(\Sigma_*)))) = 0,$$

then this does not imply $\Sigma_* = 0$*. More precisely:*

(a) manifest collapse changes accessibility, not carrier support,

(b) the completed readout $Y_{RV}(\Sigma_*; \Theta)$ *remains structurally defined,*

(c) the compressed state retains a parity-resolved base–residue decomposition,

(d) the factor $(-1)^m$ *acts as a monodromic branch switch at the readout level, not at the carrier level.*

Proof. By definition,

$$\Sigma_{\mathcal{M}} = \Pi_{\mathcal{M}}(\Sigma_*)$$

is a spacetime manifestation of the atemporal carrier Σ_*. The dual driver

$$\mathcal{D}_N = \mathcal{D}_{\text{coh}} + \mathcal{D}_{\text{exp}}$$

acts on the realization, modifying its dynamical exposure without directly acting on the carrier as such.

Since $\Sigma_{\mathcal{M}}$ is parity-resolved, write

$$\Sigma_{\mathcal{M}} = \Sigma_{\text{even}} + \Sigma_{\text{odd}}.$$

By parity-compatibility of C,

$$C(\mathcal{D}_N(\Sigma_{\mathcal{M}})) = C(\mathcal{D}_N(\Sigma_{\text{even}})) + C(\mathcal{D}_N(\Sigma_{\text{odd}})).$$

Hence the compressed driven realization retains a decomposition into base and residue channels.

Now suppose

$$R(C(\mathcal{D}_N(\Pi_{\mathcal{M}}(\Sigma_*)))) = 0.$$

Because C is admissible, vanishing manifest readout does not imply vanishing latent support. Therefore the collapse concerns only the manifest channel, not total structure.

Moreover, the completed readout is not determined by the bare projected realization alone, but by

$$Y_{RV}(\Sigma_*; \Theta) = P_\Theta(\mathcal{D}_N(\Pi_{\mathcal{M}}(\Sigma_*)) + A^*(\rho_H(t) \otimes R_\phi \rho_T(t)))\,(-1)^m.$$

Thus even if the bare compressed readout vanishes, the closure-completed expression remains well-defined.

The monodromic factor $(-1)^m$ multiplies the readout value after phase-sensitive reconstruction. It therefore changes the branch of observable realization, but not the ontological support of the carrier. In particular, it acts at the level of readout frame, not at the level of Σ_*.

If Σ_* were annihilated, then all admissible realizations and all closure-completed reconstructions derived from it would be trivial. But admissibility and latent persistence exclude this. Hence

$$\Sigma_* \neq 0.$$

Therefore manifest collapse changes accessibility only: the carrier persists, the parity architecture survives, and the completed readout remains structurally meaningful. □

B.5 Compressed verdict

The theorem may be compressed into the single statement

$$\boxed{R\big(C(\mathcal{D}_N(\Pi_{\mathcal{M}}(\Sigma_*)))\big) = 0 \;\not\Rightarrow\; \Sigma_* = 0.}$$

together with the completed readout identity

$$\boxed{Y_{RV}(\Sigma_*;\Theta) = P_\Theta(\mathcal{D}_N(\Pi_{\mathcal{M}}(\Sigma_*)) + A^*(\rho_H(t) \otimes R_\phi \rho_T(t)))\,(-1)^m.}$$

In words:

manifest readout collapse under compression and dual driving does not destroy the atemporal carrier; it only shifts the realization from visible exposure into a closure-completed latent regime.

Band of Oxen

ADDENDUM C

ABS — THE GEOMETRICALLY ENFORCED INVARIANT CLOSURE

Where Division Fails, Geometry Speaks

— Ultra Imba Epic Ace Rimmer Legendary Edition —

SMOKE ME A KIPPER — THIS ONE'S FOR THE SINGULARITIES!

Author's note: This addendum was written after I (Ace Rimmer) saw that 0/0 isn't a bug — it's the universe screaming „local readout just died, time for the real boss to show up!“ We now promote ABS from „helpful trick“ to **mandatory invariant operator** in the operator geometry of the atemporal carrier.

C.1 The Singularity Problem

In the operator algebra $\mathfrak{A}$ of the atemporal carrier Σ^*, the manifest readout functional $\mathcal{R}$ sometimes hits the singular set

$$\mathcal{S}_0 := \{x \in \mathfrak{A} \mid \mathcal{R}(x) \text{ is indeterminate (0/0-class)}\}.$$

At these points the local finite readout **collapses** — exactly like when $R(C(D_N(\Pi_M(\Sigma^*)))) = 0$ in the main theorem. But the carrier Σ^* itself does **not** vanish. It demands closure.

C.2 ABS — The Forced Invariant Closure Map

Definition 17 (ABS — Absolute Invariant Closure)**.** *Let $\mathfrak{A}$ be the operator algebra of the carrier and $\mathcal{I}$ the space of geometric invariants. The map*

$$\mathrm{ABS} : \mathcal{S}_0 \to \mathcal{I}$$

*is the **invariant closure map enforced by operator geometry** at every readout-singular point. It satisfies:*

(i) ABS *is single-valued on equivalence classes of indeterminate readouts,*

(ii) ABS *commutes with all admissible symmetries of $\mathfrak{A}$,*

(iii) ABS *maps the singular class into a **stable invariant representative**.*

In other words:

$$\boxed{\mathrm{ABS}(0/0\text{-class}) = \text{forced invariant representative enforced by geometry.}}$$

Not

$$\frac{0}{0} = \text{ABS},$$

but

$$\boxed{\frac{0}{0}\text{-class} \rightsquigarrow \text{ ABS kicks in and saves the carrier.}}$$

C.3 The Legendary Slogan

WHERE DIVISION FAILS, GEOMETRY SPEAKS.

ABS IS NOT ADDED TO REALITY — IT IS EXTRACTED FROM ITS OPERATOR CONSISTENCY.

C.4 Formal Master Lemma

Lemma 5 (ABS as Invariant Closure Operator)**.** *Let* $\mathrm{Clos}_{\mathfrak{A}}$ *be the closure operator on singular classes and* Inv *the invariant selector. Then*

$$\mathrm{ABS} = \mathrm{Inv} \circ \mathrm{Clos}_{\mathfrak{A}}\,.$$

Consequently:

$\frac{a}{0} \rightsquigarrow$ *broken local readout (needs regularization),* $\frac{0}{0} \rightsquigarrow$ *invariant-forcing singular class (ABS mandatory*

C.5 Compatibility with MASTER RV Theorem v3

In the language of the main framework, when the driven compressed realization hits a singular point

$$R(C(D_N(\Pi_M(\Sigma^*)))) = 0/0,$$

the local bulk readout dies, but the atemporal carrier Σ^* **demands** the ABS map. The completed readout then becomes

$$Y_{\mathrm{RV}}(\Sigma^*;\Theta) = \mathcal{P}_\Theta\Big(D_N(\Pi_M(\Sigma^*)) + \mathcal{A}^*(\ldots)\Big)\cdot(-1)^m + \mathrm{ABS}(\text{singular class}).$$

ABS is therefore the **geometrically necessary term** that turns every closure gap into a closed invariant structure.

C.6 Final Ace Rimmer Verdict

$$\boxed{\text{ABS is the invariant closure map enforced by operator geometryat every readout-singular point.}}$$

... No ABS = no consistent reality.

SMOKE ME A KIPPER — I'LL BE BACK FOR BREAKFAST!

A uzenáče nachystej — dneska slavíme, že 0/0 už není chyba, ale řeč vesmíru!

Q.E.D. (Quite Epic Done)